

Performance Analysis of MPPT Techniques for Photovoltaic Systems Under Partial Shading Conditions

Arslan Khalid, Saad Mekhilef, Mehdi Seyedmahmoudian, Alex Stojcevski
School of Science, Computing and Engineering Technologies,
Swinburne University of Technology,
Melbourne, Australia
{akhalid, smekhilef, mseyedmahmoudian, alex.stojcevski}@swin.edu.au

Sajib Ahmed
Faculty of Engineering and IT,
University of Technology Sydney,
Sydney, Australia
sajib.ahmed@uts.edu.au

Abstract—Solar energy is one of the most promising renewable resources due to its abundance, sustainability, and eco-friendly nature. The efficiency of photovoltaic (PV) systems largely depends on temperature and solar irradiance. Under uniform conditions, the current-voltage (I-V) curve exhibits a single maximum power point (MPP), which can be tracked using Maximum Power Point Tracking (MPPT) techniques. Among these, the Perturb and Observe (P&O) algorithm is widely adopted for its simplicity and low cost. However, under partial shading conditions (PSC), multiple peaks appear on the power-voltage (P-V) curve, causing conventional P&O to track local maxima instead of the global MPP, resulting in significant power loss. To address these limitations, this study presents a systematic comparison of the traditional P&O method with three enhanced variants, designed to improve convergence speed, tracking accuracy, and adaptability under dynamic shading scenarios. MATLAB/Simulink simulations are carried out under identical conditions to evaluate their performance in terms of efficiency, response time, and stability. The results demonstrate that the proposed enhanced methods offer faster convergence and higher energy harvesting efficiency compared to the conventional approach, providing practical insights for selecting suitable MPPT strategies for real-world PV systems.

Index Terms—Differential Evolution (DE), Global Maximum Power Point (GMPP), Maximum Power Point Tracking (MPPT), Partial Shading Condition (PSC), Particle Swarm Optimization (PSO), Perturb and Observe (P&O), Photovoltaic (PV)

I. INTRODUCTION

The paper addresses the challenge of Maximum Power Point Tracking (MPPT) under partial shading conditions (PSC), where conventional methods often fail due to multiple local maxima on the power-voltage (P-V) curve. Unlike uniform irradiance, where a single MPP exists, PSC complicates tracking and reduces energy harvesting efficiency. Traditional algorithms like Perturb and Observe (P&O) are simple and low-cost but prone to locking onto local peaks [1]. Heuristic methods such as Differential Evolution (DE) and Particle Swarm Optimization (PSO) improve global search but often converge slowly and inconsistently under dynamic shading [2]. Moreover, these methods generally require higher computational resources, which can limit their applicability in low-cost or embedded PV systems.

This study presents a comparative analysis of P&O, DE, PSO, and the Trapezium Method using MATLAB/Simulink under various partial shading scenarios. The Trapezium Method deterministically segments the P-V curve and evaluates the area under each segment to identify the most promising region. This geometric approach avoids false maxima, reduces computational complexity, and ensures reliable GMPP detection suitable for embedded systems [3] - [5]. Simulation results show that it consistently outperforms traditional and heuristic methods in tracking accuracy, convergence speed, and robustness.

The main contributions include a thorough performance comparison of classical, heuristic, and deterministic MPPT techniques under PSC, validation of the Trapezium Method as a low-computation and reliable solution for embedded PV systems, and demonstration of its superior performance in dynamic shading conditions. The study also provides insights into the trade-offs between computational cost and energy harvesting efficiency, laying the groundwork for future hybrid or adaptive MPPT strategies.

The paper is organized as follows: Section I reviews the operational principles of the four MPPT algorithms. Section III presents comparative simulation results, analyzing their effectiveness and limitations. Section IV concludes with key findings and practical insights for real-world PV system design.

II. ANALYSIS OF MPPT ALGORITHM OPERATIONS

To optimize the MPPT under demanding conditions like PSC, conventional approaches require refinement. This study examines the P&O method alongside two advanced adaptations designed to overcome its shortcomings. These improved algorithms enhance convergence speed, minimize steady-state fluctuations, and effectively locate the GMPP. Executed through a direct control approach employing a buck-boost converter to dynamically adjust the operating voltage, the operational principles, strengths, and limitations of all four methods are thoroughly detailed in the sections below.

1) Perturb and Observe (P&O) Algorithm

The Perturb and Observe (P&O) algorithm is a popular method for tracking the maximum power point (MPP) in photovoltaic (PV) systems. It functions by slightly modifying the voltage and analyzing the resulting variation in power output to find MPP [1]. The algorithm determines the difference in power and voltage using $\Delta P = P_{\text{new}} - P_{\text{old}}$ and $\Delta V = V_{\text{new}} - V_{\text{old}}$, where P_{new} and P_{old} are the current and previous power values, and V_{new} , V_{old} are the corresponding voltage readings. If ΔP is positive, the duty cycle D_{old} is increased by a small fixed increment ΔD ; otherwise, it is decreased. This approach enables the system to continuously adjust the operating point, updating the duty cycle D_{new} to move closer to the MPP. The flowchart in Fig. 1 illustrates the process, with PV module characteristics are detailed in TABLE I.

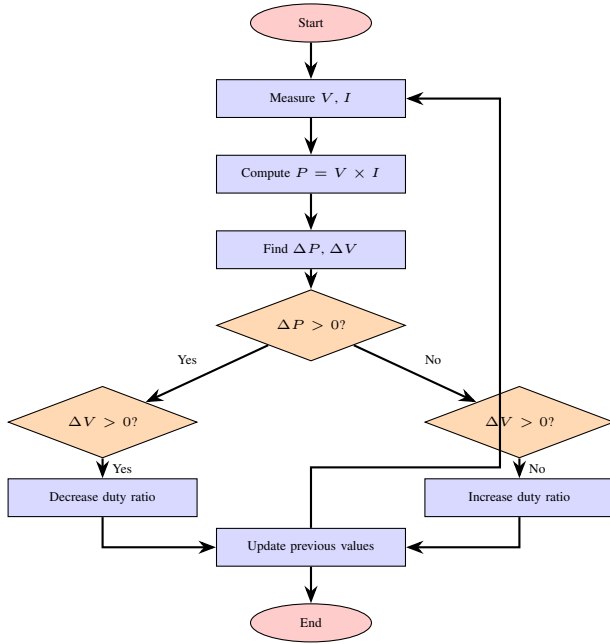


Fig. 1: Flowchart diagram of P&O algorithm

2) Differential Evolution Algorithm (Modified)

The DE algorithm, introduced in [3] enhances MPPT performance in photovoltaic systems by employing three main operations: mutation, crossover, and selection. Initially, a donor vector $\mathbf{v}_i^{(g)}$ is formed by combining three randomly selected individuals from the current population, scaled by a factor F . Then, a trial vector element $u_{i,j}^{(g)}$ is created by mixing components of the donor and target vectors based on a predefined crossover probability C_r . Finally, the selection process compares the output power represented by the fitness function of the trial vector with the current one. If the trial yields better performance, it replaces the existing solution, becoming $\mathbf{x}_i^{(g+1)}$ in the next generation. where, the term $\mathbf{x}_i^{(g+1)}$ refers to the updated solution (vector) for the i -th individual in the next generation ($g + 1$). Fig. 2 shows a flowchart outlining the step-by-step process of the proposed approach for clear visualization.

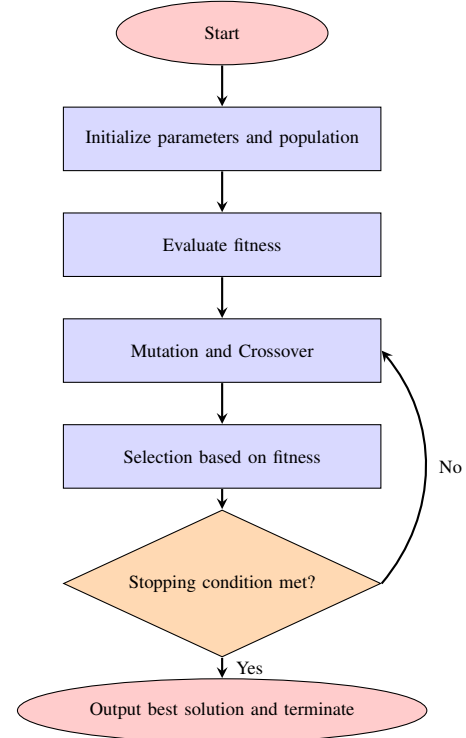


Fig. 2: Flowchart diagram of DE algorithm

3) Particle Swarm Optimization Algorithm (Modified)

The PSO technique, outlined in [5], is applied to improve MPPT in PV arrays, especially when partial shading is present. This algorithm models a group of particles that explore the search space by adjusting their velocities and positions. Each particle considers both its historically best location and the overall best position identified by the group. The velocity update uses parameters including inertia weight w , acceleration coefficients c_1 and c_2 , and random values r_1 and r_2 , which help balance exploration and exploitation. The new particle position is determined by adding the updated velocity to the current position. The fitness value of a particle, represented as $f(\mathbf{x}_i) = P(\mathbf{x}_i)$, is used to assess performance in terms of power output. The term $\mathbf{x}_i$ denotes the present location or candidate solution of the i -th particle within the search domain. The iterative behavior of the particles allows convergence toward the GMPP. A schematic flow of the algorithm's procedure is provided in Fig. 3 to illustrate the process.

4) Trapezium Method (Modified)

The algorithm is inspired by the Maximum Power Tracking (MPT) method described in [6] and employs a numerical integration technique to estimate the area under a curve. This method divides the interval $[a, b]$ into n equal segments, each of width $h = \frac{b-a}{n}$. By evaluating the continuous function $f(x)$ at these discrete points, the algorithm approximates the integral by summing the areas of trapezoids formed under the curve. Specifically, the integral is approximated by adding the function values at the interval endpoints and twice the sum

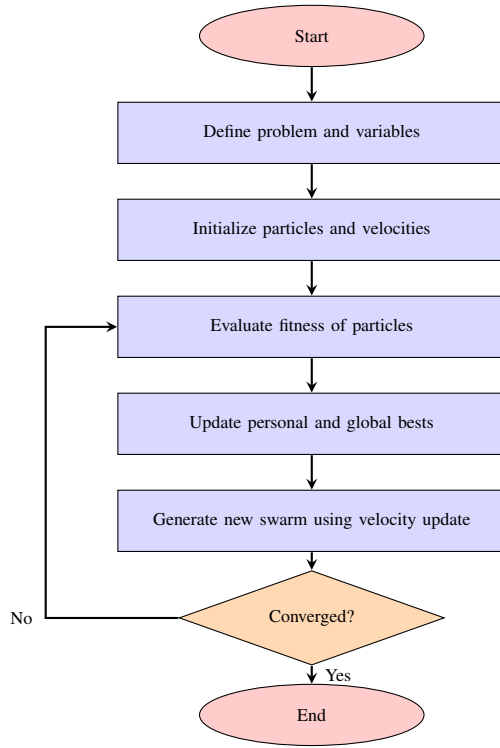


Fig. 3: Flowchart of PSO Algorithm

TABLE I. PHOTOVOLTAIC MODULE ELECTRICAL CHARACTERISTICS

Category	Parameter	Value	Unit	Notes
Module Ratings				
1	Maximum Power Output	49.9984	W	At STC
2	Number of Cells (N_{cell})	60	–	Series-connected
Electrical Parameters (STC)				
3	Open-Circuit Voltage (V_{oc})	9.075	V	–
4	Short-Circuit Current (I_{sc})	7.84	A	–
5	Voltage at MPP (V_{mp})	6.8025	V	–
6	Current at MPP (I_{mp})	7.35	A	–
Temperature Coefficients				
7	V_{oc} Coefficient	-0.36099	%/°C	Negative temp. drift
8	I_{sc} Coefficient	+0.102	%/°C	Positive temp. drift

of the function values at the internal points $x_k = a + kh$, where k ranges from 1 to $n - 1$, and then multiplying by half the segment width. This approach simplifies the integral calculation by approximating the area with trapezoids rather than rectangles, improving accuracy especially for smooth functions. The flowchart in Fig. 4 outlines the algorithm's steps, highlighting its logical progression and key decision points. Overall, this structured numerical method provides a reliable and efficient way to approximate definite integrals necessary for the tracking algorithm.

III. COMPARATIVE SIMULATION RESULT

Three partial shading conditions (PSC-A, PSC-B, and PSC-C) were designed to position the global maximum power point (GMPP) at different locations on the P–V curve:

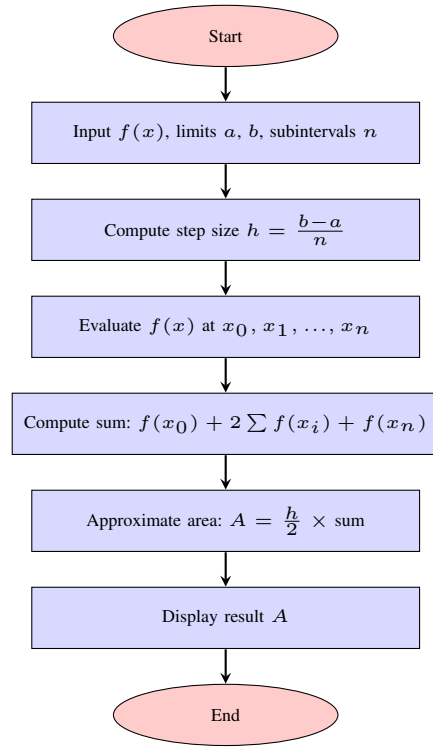


Fig. 4: Process Flow of the Trapezium MPPT Approach

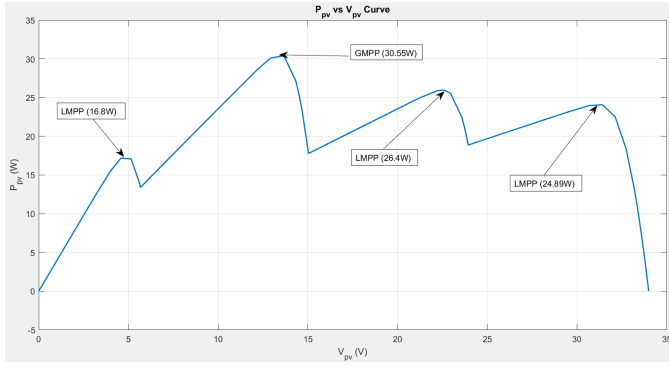
TABLE II. PERFORMANCE COMPARISON OF MPPT ALGORITHMS UNDER PARTIAL SHADING

Partial Shading Case	P&O	DE	PSO	Trapezium Method
Tracking Time (s)				
PSC-A	0.12	0.07	0.32	0.28
PSC-B	0.18	0.11	0.18	0.09
PSC-C	0.38	0.09	0.10	0.31
Mean Output Power (W)				
PSC-A	24.12	24.14	30.55	30.55
PSC-B	37.13	37.14	52.08	52.83
PSC-C	72.66	72.66	72.66	72.66
GMPP Tracked				
PSC-A	No	No	Yes	Yes
PSC-B	No	No	No	Yes
PSC-C	Yes	Yes	Yes	Yes

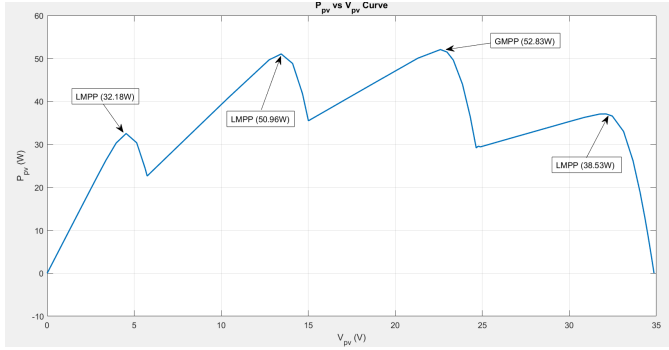
PSC-A (Left): 150, 300, 500, 100 W/m²; PSC-B (Center): 1000, 300, 500, 150 W/m²; PSC-C (Right): 400, 700, 1000, 300 W/m².

All tests were performed at a constant temperature of 25°C, using a fixed duty cycle step size ($\Delta d = 0.001$) and a sampling time of 0.005 s. The corresponding shading patterns are shown in Fig. 5.

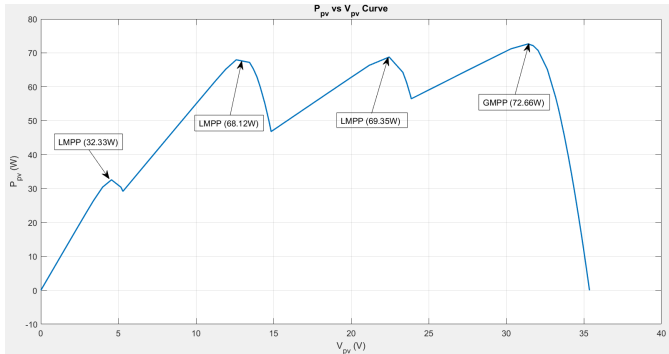
Four MPPT algorithms namely P&O, DE, PSO and the Trapezium Method were evaluated in MATLAB/SIMULINK. P&O demonstrated fast tracking but failed to reach the GMPP in two cases, resulting in lower power output despite minimal oscillations. DE showed the fastest convergence and low oscil-



(a) PSC-A



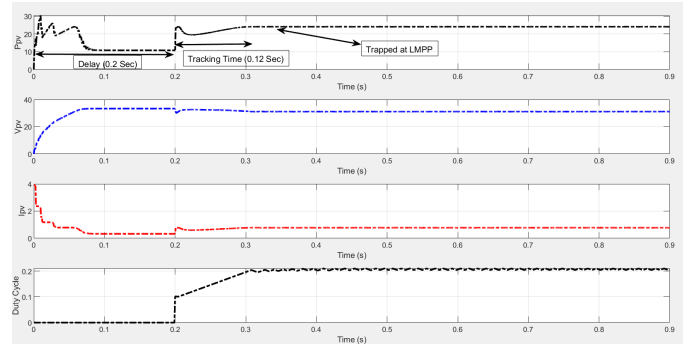
(b) PSC-B



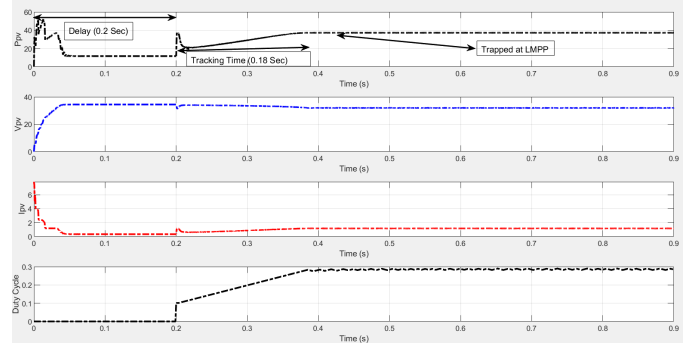
(c) PSC-C

Fig. 5: MPPT performance under different partial shading conditions: (a) PSC-A, (b) PSC-B, (c) PSC-C.

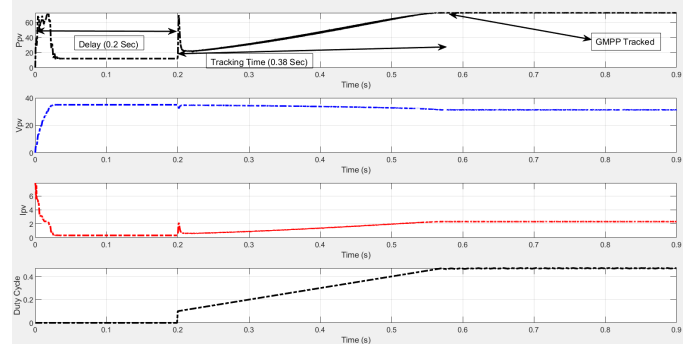
lations but lacked consistency in GMPP detection. Meanwhile, the PSO algorithm produced moderate power but experienced higher oscillations and slower tracking, correctly identifying the GMPP only in two cases. In contrast, the Trapezium Method consistently located the GMPP across all cases, yielding the highest power output with low oscillations and a good balance of speed and accuracy. Performance comparisons are presented in TABLE II, and Figs. 6–9 illustrate the dynamic responses under varying shading conditions.



(a) PSC-A



(b) PSC-B



(c) PSC-C

Fig. 6: MPPT behavior of the P&O method under different partial shading scenarios: (a) PSC-A, (b) PSC-B, and (c) PSC-C.

IV. CONCLUSION

This study compared four MPPT algorithms, namely P&O, PSO, DE, and the Trapezium Method, under various partial shading scenarios. The Trapezium Method consistently tracked the global maximum power point (GMPP) with high accuracy, minimal oscillations, and strong robustness, outperforming the others in reliability and stability. While DE achieved the fastest convergence, it occasionally settled on local maxima. PSO exhibited moderate performance with higher oscillations and slower response, whereas conventional P&O struggled under complex shading. These findings emphasize the contribution of the Trapezium Method as an efficient and deterministic MPPT solution, capable of delivering superior performance without the computational complexity of metaheuristic algorithms.

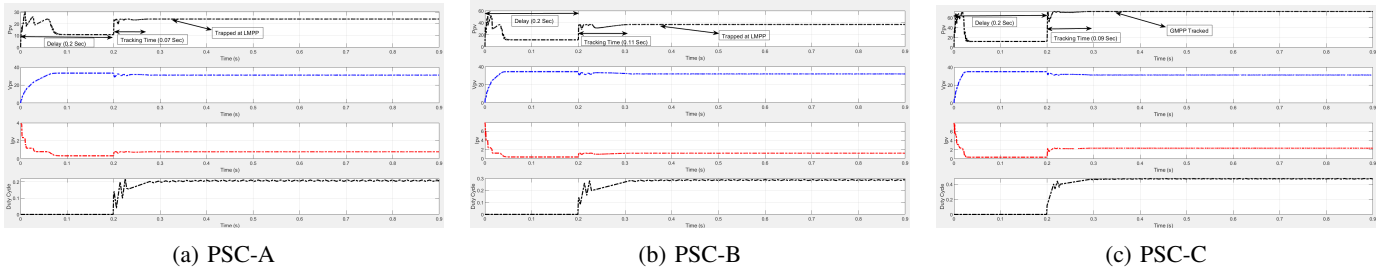


Fig. 7: Tracking performance of the DE Method under three unique partial shading scenarios: (a) PSC-A, (b) PSC-B, and (c) PSC-C.

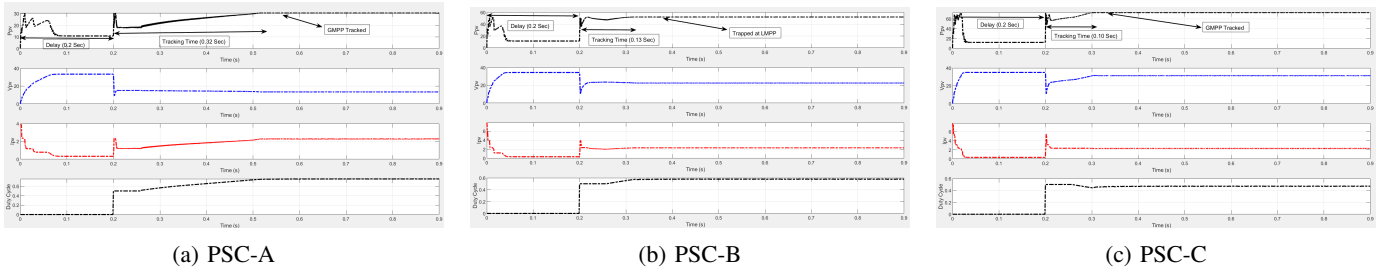


Fig. 8: Performance analysis of the PSO technique under varying shading scenarios: (a) PSC-A, (b) PSC-B, and (c) PSC-C.

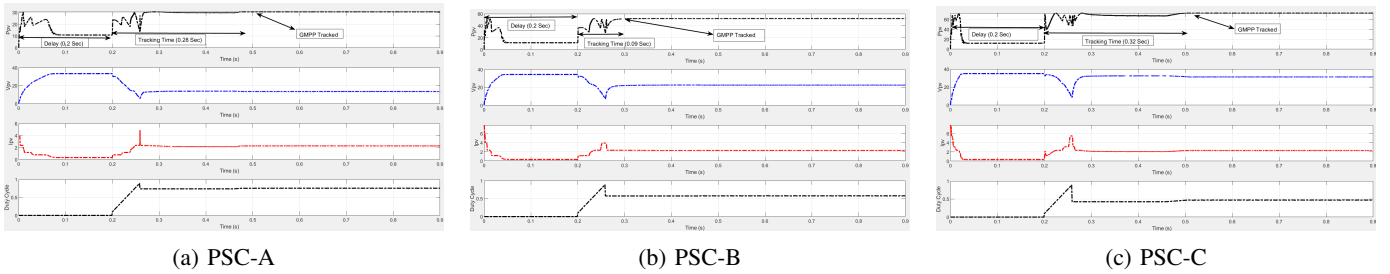


Fig. 9: Observed performance curves of the proposed Trapezium MPPT technique under varying irradiance configurations: (a) PSC-A, (b) PSC-B, and (c) PSC-C.

Overall, the results demonstrate its practical applicability for real-world PV systems, ensuring enhanced energy harvesting even under challenging shading conditions.

Future work will investigate adaptive enhancements to further accelerate convergence during rapidly changing conditions, integration of predictive control to anticipate shading effects, and optimization for large-scale PV arrays in grid-connected environments. Additionally, hybrid frameworks combining the geometric principles of the Trapezium Method with global search techniques will be explored to improve performance in highly dynamic and partially shaded scenarios. Further studies will also focus on experimental validation using hardware prototypes to confirm simulation outcomes and assess real-time operational feasibility.

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